JALA UNIVERSITY

University Library SPA

Modern Library Management & Catalog Single-Page Application

Capstone Project Defense — Web Development (CSWB-366)

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- **Live Demo:** https://d1376jrski7xuw.cloudfront.net

The Problem & Motivation

Academic Library Challenges

- **Scattered Information:** Students struggle to find whether physical copies are available on campus or checked out.
- **Outdated UI/UX:** Legacy university library systems have poor mobile support and cumbersome workflows.
- **Queue Transparency:** Lack of visibility into waitlists when high-demand books are unavailable.

The Opportunity

Create a **lightning-fast, mobile-first web application** that integrates public bibliographic databases with live campus physical copy inventory and smart lending rules.

The Solution: University Library SPA

Key Value Propositions

1. **Global Search + Local Inventory:** Combines **Open Library's** 30M+ book catalog with a local **PostgreSQL** physical copy availability tracker.
2. **Automated Lending Lifecycle:** 14-day loans, due-date calculation, renewal validation (+7 days), and overdue tracking.
3. **Fair FIFO Reservation Queue:** Automatic hold placement when 0 copies exist, with a 48-hour exclusive pickup window.
4. **Personal Bookshelf:** Track current reading (`READING`), celebrate finished books (`COMPLETED`), and view loan history.

Cloud-Native Architecture (AWS)

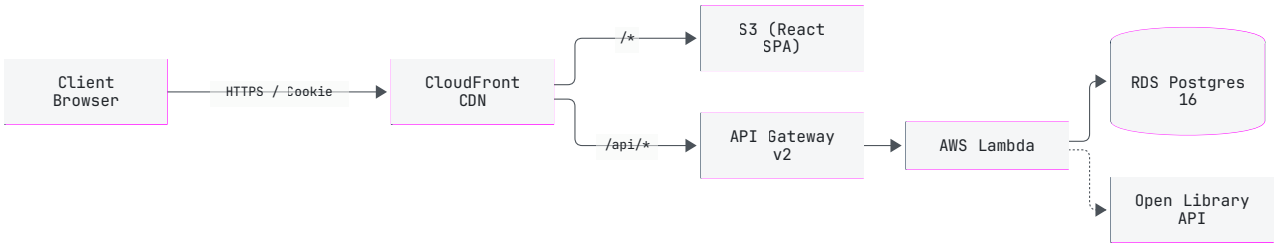

Figura 1 — Diagrama

- **Unified Domain:** `d1376jrski7xuw.cloudfront.net` (Zero cross-site cookie blocking).
- **Strict Free Tier:** `db.t4g.micro`, 20 GB gp3, Single-AZ ($0.00 infrastructure cost).

Core Business Logic & Concurrency

Rigorous Lending Rules (BR-01 through BR-12)

- **BR-01 to BR-05 (Borrowing):** Max 5 active loans per user. Loan period: 14 days. Copies decremented atomically in database transactions.
- **BR-08 & BR-09 (Renewals):** Capped at 2 renewals (+7 days each). Denied if overdue or if another student is waiting in the reservation queue.
- **BR-10 (FIFO Queue):** When available copies = 0, users join an orderly queue.
- **BR-11 & BR-12 (48h Hold Window):** When a book is returned, it is held exclusively for user #1 for 48 hours.

Mobile-First Responsive Design

Optimized for Every Screen Size

- **Accessible Navigation:** Slide-out mobile navigation drawer with touch targets ≥44px\ge 44\text{px}.
- **Adaptive Catalog Grid:** 1 column on small phones (< 440px) →\to 2 on phablets →\to 3 on tablets →\to 4 on desktop.
- **Ergonomic Controls:** Full-width segmented controls and horizontal swipeable category chips.
- **Dark & Light Modes:** Instant CSS variable toggling with persistent theme memory.

Software Quality & Testing

Test Suite Summary: 161 Tests Passing (57 Suites)

- **Frontend (47 Suites / 130 Tests):**
  - Component rendering, loading skeletons, and error fallbacks.
  - User interaction flows (Search, Filter, Borrow, Return, Renew, Place Hold).
  - Custom hook tests (`useCatalogSearch`, `useMyLoans`, `useReservations`, `useReadingStatus`).
  - Mobile menu toggle accessibility tests.
- **Backend (10 Suites / 31 Tests):**
  - Business rules validation in isolated service layers.

    - Concurrency tests: race conditions on the last available physical copy.
    - Authentication middleware, JWT verification, and bcrypt hashing.
- **TypeScript:** 100% strict type checking with **zero** `any` in new code.

Infrastructure as Code (Terraform)

Why IaC with Terraform?

- **100% Reproducible:** The entire AWS environment (S3, CloudFront, Lambda, API Gateway, RDS, Security Groups) is defined in `infra/*.tf`.
- **Zero Cost Surprise Protection:** Hardcoded Free Tier limits:
    - `instance_class = "db.t4g.micro"`
    - `allocated_storage = 20`
    - `max_allocated_storage = 0` (No storage autoscaling)
    - `multi_az = false`
- **Reverse Proxy Routing:** CloudFront handles both frontend assets and backend API proxying under a single certificate.

Live Demonstration Flow (6 Minutes)

1. **Catalog Search & Real-Time Availability:** Search for "Tolkien", filter by "Fiction", observe copy count pips (3/3).
2. **Mobile-First Experience:** Switch to mobile viewport: open drawer menu, browse categories with touch gestures.
3. **Borrowing a Book:** Authenticate, click Borrow, verify automatic due date calculation (+14 days).
4. **Renewal Validation:** Click Renew in "My Loans" (+7 days extension, counter increments to 1/2).
5. **FIFO Reservation Queue:** Find a book with 0 copies, click "Place Hold", observe queue position #1.
6. **Return & Reading Tracker:** Return borrowed book, move to "Reading Log", celebrate completed title.

Conclusions & Key Learnings

What Was Achieved:

- Complete end-to-end fullstack platform meeting all IEEE 29148 SRS requirements.
- Modern serverless AWS architecture deployed through Terraform.

- Solved complex cross-site authentication cookie issues using CloudFront Reverse Proxy.
- Comprehensive test coverage with 161 automated tests.

Thank You!

- **Live Application:** https://d1376jrski7xuw.cloudfront.net
- **Repository:** https://github.com/Catriel42/university-library
- **Questions & Discussion**